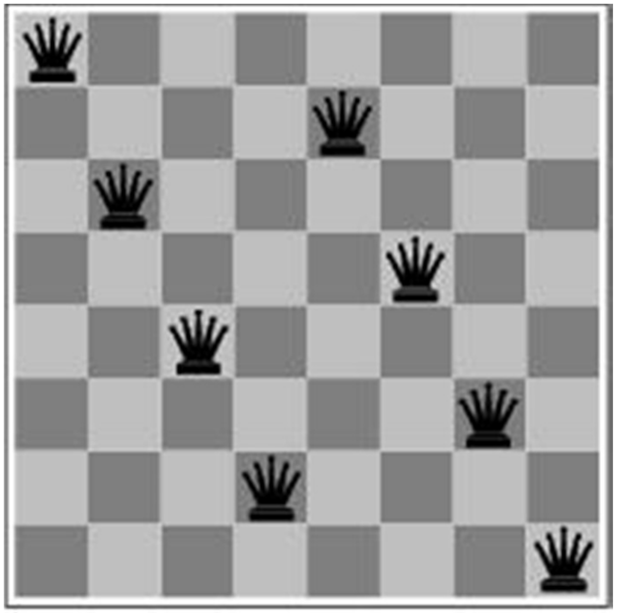
Local Search and Optimization Algorithms

Outline

- Local search techniques and optimization
  - Hill-climbing
  - Simulated annealing
  - Genetic algorithms
  - Issues with local search

Local search and optimization

- Previous Algorithms: path to goal is solution to problem
  - systematic exploration of search space.
- This lecture: a state is solution to problem
  - for some problems path is irrelevant.
  - E.g., 8-queens
- Different algorithms can be used
  - Like Local search



Goal Satisfaction

reach the goal node
Constraint satisfaction

Optimization

optimize(objective fn)
Constraint Optimization

You can go back and forth between the two problems
Typically in the same complexity class

Local search and optimization

- Local search
  - Keep track of single current state
  - Move only to neighboring states
  - Ignore paths

- Advantages:
  - Use very little memory
  - Can often find reasonable solutions in large or infinite (continuous) state spaces.

- "Pure optimization" problems
  - All states have an objective function
  - Goal is to find state with max (or min) objective value

Hill-climbing (Greedy Local Search) max version

```
function HILL-CLIMBING( problem) return a state that is a local maximum
    input: problem, a problem
    local variables: current, a node.
                     neighbor, a node.

    current ← MAKE-NODE(INITIAL-STATE[problem])
    loop do
        neighbor ← a highest valued successor of current
        if VALUE [neighbor] ≤ VALUE[current] then return STATE[current]
        current ← neighbor
```

min version will reverse inequalities and look for lowest valued successor

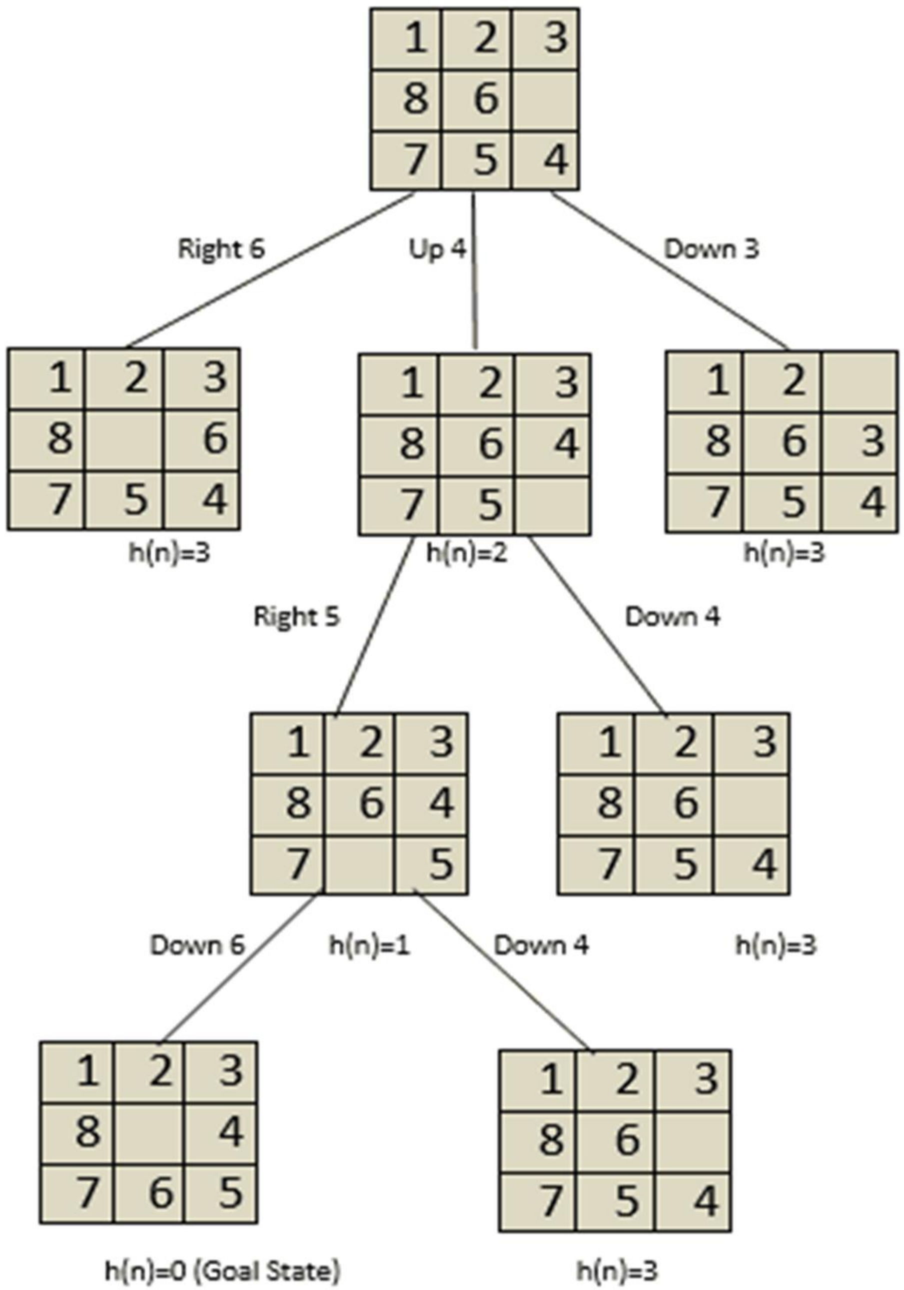
Hill-climbing search

- "a loop that continuously moves towards increasing value"
  - terminates when a peak is reached
  - Also known as greedy local search
- Value can be either
  - Objective function value
  - Heuristic function value (minimized)

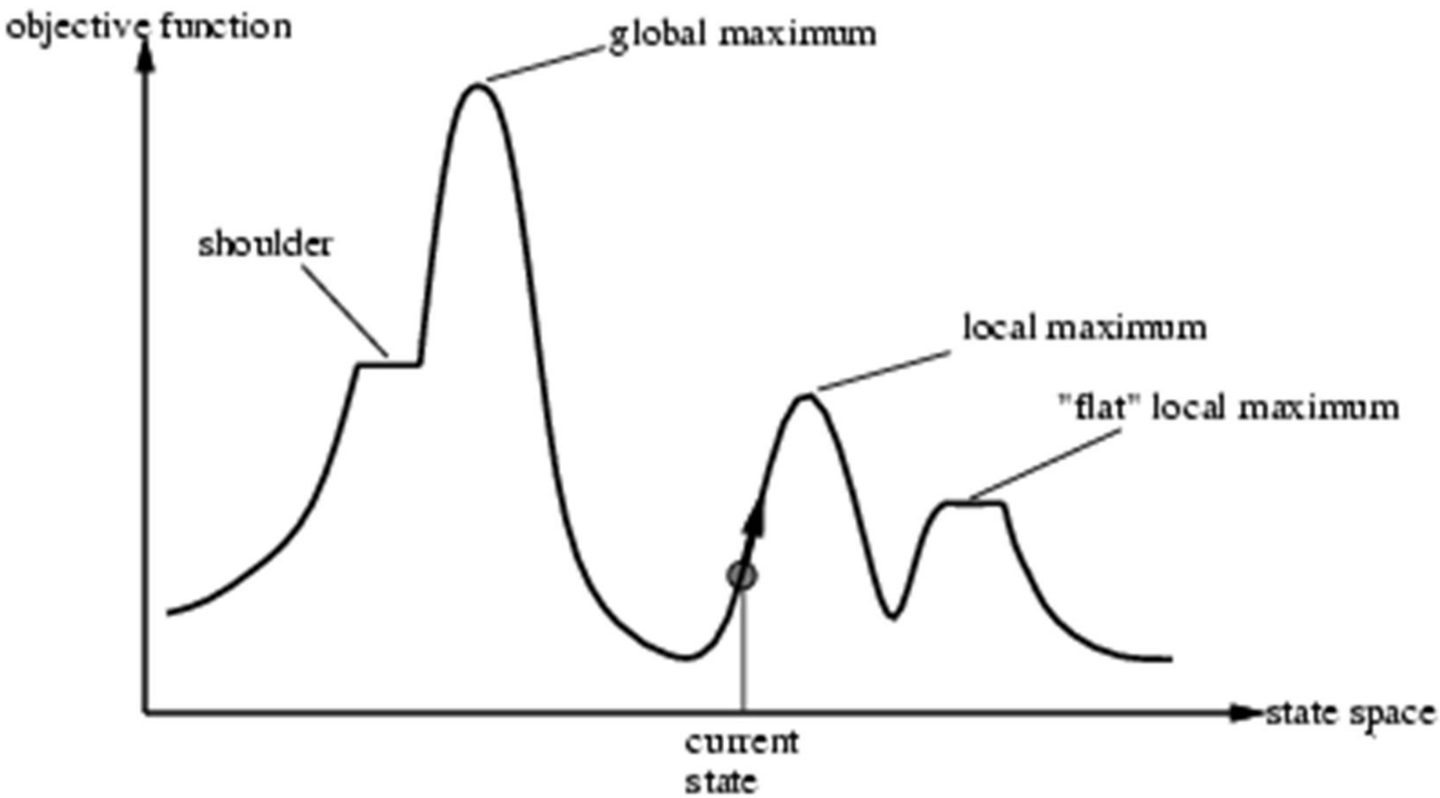
- Hill climbing does not look ahead of the immediate neighbors
- Can randomly choose among the set of best successors
  - if multiple have the best value

- "Climbing Mount Everest in a thick fog with amnesia"

Working of Hill Climbing Search Algorithm



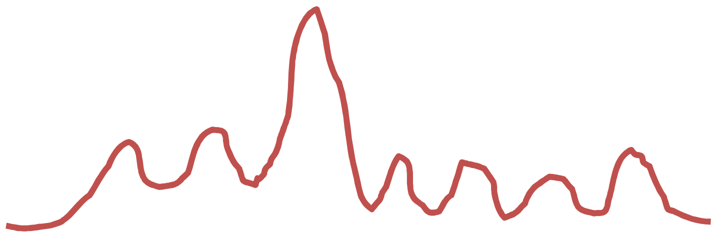
“Landscape” of search



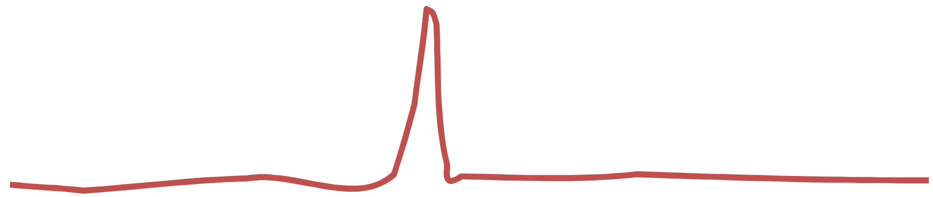
Hill Climbing gets stuck in local minima

Hill Climbing Drawbacks

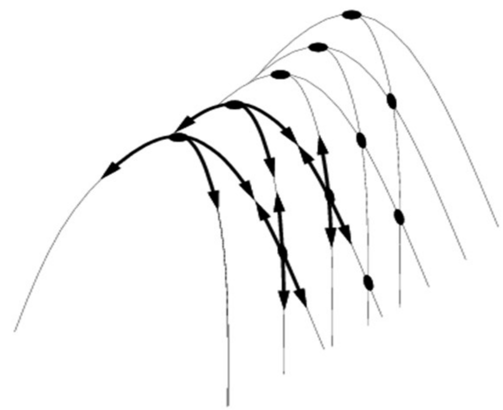
- Local maxima



- Plateaus



- Diagonal ridges



Hill-climbing: stochastic variations

- Stochastic hill-climbing
  - Random selection among the uphill moves.
  - The selection probability can vary with the steepness of the uphill move.

- To avoid getting stuck in local minima
  - Random-walk hill-climbing
  - Random-restart hill-climbing
  - Hill-climbing with both